Somesh Kennedy

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EDUCATION

Master of Science in Computer Science August 2024 - May 2026
University of Texas - Dallas (Richardson, TX) GPA: 3.97/4.0
Courses: *Artificial Intelligence, Natural Language Processing, Big Data Management, Database Designs, Machine Learning*

Bachelor of Technology in Computer Science and Engineering December 2020 - June 2024
National Institute of Technology Puducherry (Karaikal, India) GPA: 8.24/10.0

TECHNICAL SKILLS

Programming: Python, C++, HTML/CSS, SQL, Bash/Shell, Git, Data Structures
AI/ML: LLMs, Agentic AI, GenAI, Prompt Engineering, NLP, RAG, MCP, Model Optimization, LSTMs, Feature Engineering
Frameworks & Libraries: LangChain, LangGraph, CrewAI, AutoGen, Pandas, Numpy, Tensorflow, Keras, PyTorch
Cloud/Backend: AWS, PostgreSQL, MongoDB, ETL/ELT, FAISS, Chroma, Docker, Kubernetes
Web/Tools: Streamlit, Django, FASTAPI, Claude Code, Codex, Github, HuggingFace, VS Code, Vertex AI, Tableau

EXPERIENCE

AI Engineer Intern **June 2025 - August 2025**
WorldLink US *Frisco, Texas*
- Architected risk assessment agents using **OpenAI**, **RAG**, and **PostgreSQL**, increasing audit success rate by **12%**.
- Orchestrated **agentic pipelines** via **LangGraph** and **MCP** for runtime tool discovery, reducing execution steps by **35%.**
- Decoupled agent workflows into **Python microservices** using **FastAPI, Docker containers** and **AWS EC2**.
- Automated governance **workflow triggers** using **AWS Lambda**, accelerating compliance processing by **45%**.
- Improved **LLM validation cycles** by automating **prompt evaluation benchmarks** using **PyTest** and **MLflow**.

Software Engineer Intern **January 2024 - March 2024**
P9 Technologies Pvt Ltd *Puducherry, India*
- Designed a **multi-threaded LLM integration layer** with rate-limiting, boosting system throughput to **3x** peak load.
- Engineered **Python CI/CD pipelines** to validate **data lineage** and **model quality**, cutting manual QA effort by **15%**.
- Boosted retrieval precision by **25%** by constructing **hybrid vector search pipelines** with **FAISS** and **PostgreSQL**.
- Developed a **LangChain** chatbot integrated with enterprise databases, improving search response accuracy by **18%**.
- Wrote unit tests using **PyTest** to cover core backend features, reducing code bugs by **20%**.

Data Science Intern **May 2023 - July 2023**
NIT Puducherry *Karaikal, India*
- Engineered ensemble **XGBoost** and **PyTorch CNN** architecture for predictive modeling, achieving **94%** accuracy.
- Built chemical detection models utilizing **MobileNetV2 recognition pipelines**, boosting detection accuracy by **14%**.
- Implemented **data cleaning** algorithms using **PySpark**, reducing raw signal noise from time-series data by **20%**.
- Containerized **ML processing pipelines** using **Docker** and streamlined environment setup across **9,000+** data samples.

PROJECTS

Agentic Customer Support Platform | Agentic AI, PostgreSQL, Django
- Architected a **multi-agent** customer support system using **CrewAI** and **MCP** to handle autonomous reasoning flows.
- Designed custom prompt templates that autonomously resolved **72%** of support queries end-to-end.
- Developed a **Django** front-end application to capture user queries, reducing human escalation by 40%.

RAG Enterprise Knowledge Agent | Agentic AI, Pinecone, RAG, FASTAPI
- Constructed a production **RAG** system using **LangChain** and **Pinecone** to manage semantic document retrieval.
- Engineered automated data pipelines to index **50+** enterprise documents, achieving an **88%** answer relevance score.
- Exposed retrieval workflows via a **FASTAPI backend application** to easily query the underlying knowledge base.

Credit Card Fraud Detection System | ML, Kaggle, Feature Engineering
- Built an end-to-end fraud detection pipeline on a Kaggle dataset of **280K+** transactions using **EDA, SMOTE** for class balancing, and feature engineering on **30** features.

Autonomous AI Research Assistant | LLMs, API Integration, Streamlit
- Built an intelligent research assistant using **CrewAI**, **Gemini** and **Streamlit** to automate end-to-end literature reviews.
- Integrated the **ArXiv API** to dynamically retrieve, cluster, and summarize over **100** research papers.

PUBLICATIONS

- **Somesh K, C. Shanmukha Srinivas et al.,** *Tea Leaf Disease Classification with Ensemble Stacking,* ICDLAIR 2024 (IEEE)
- **Harish S, Somesh K et al.,** *FinSecure: Utilizing IoT Sensors for Formaldehyde Detection,* IConSCEPT 2024 (IEEE)